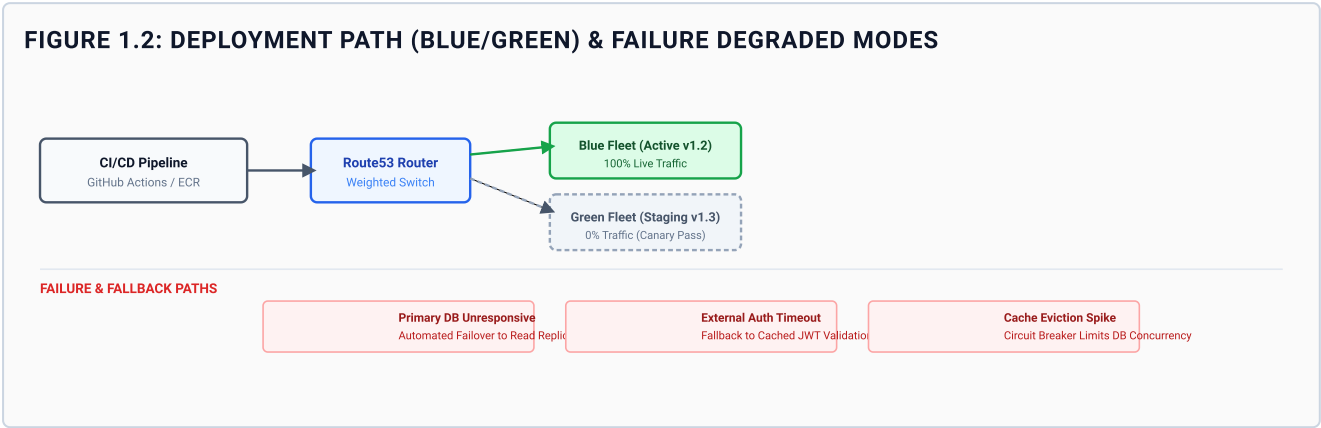
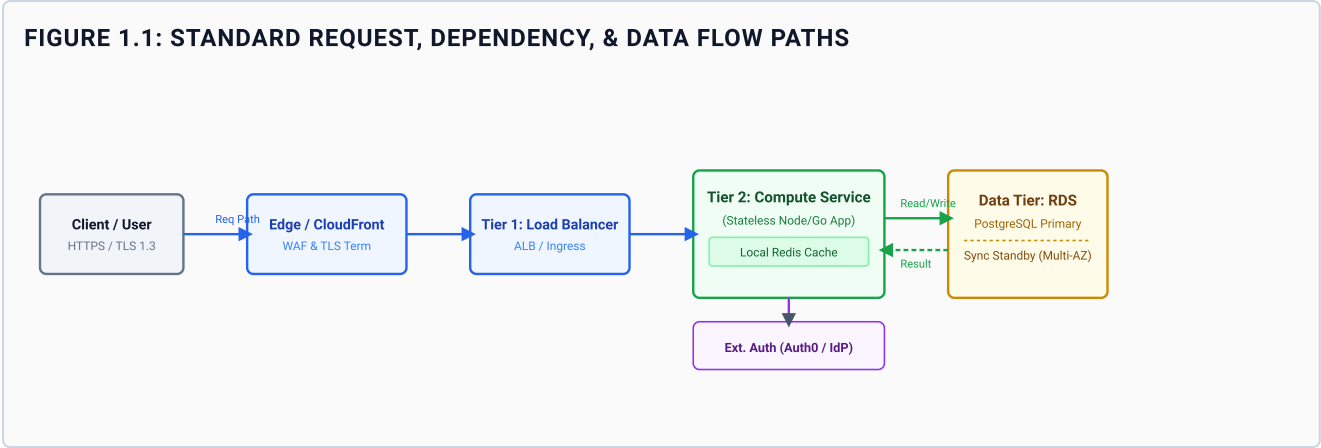


Two-Tier Service Architecture & SRE Manual

PRODUCTION OPERATIONS, SLO FRAMEWORK, COST MODEL & ARCHITECTURE DECISIONS

Target System:	Web Application & Relational DB Tier	Document ID:	ARCH-OPS-2T-2026	Revision:	v2.4	Scope:	Production Engineering & FinOps
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1. SYSTEM FLOW & OPERATIONAL PATH VISUALIZATIONS



Primary Request & Data Path Clients establish TLS 1.3 encrypted sessions via CloudFront edge nodes. Edge WAF inspects and routes valid requests to the Tier 1 Application Load Balancer (ALB). The ALB distributes traffic evenly across stateless Tier 2 application nodes running in AWS ECS Fargate. Node application instances query the local Redis cache layer for session and read data; cache misses execute parameterized SQL queries against the primary RDS PostgreSQL database using connection pooling (PgBouncer).	Deployment & Failure Path Protocols Deployments follow a zero-downtime Blue/Green strategy. CI/CD provisions Green tasks, runs health checks, and shifts ALB target group weights incrementally (10% -> 50% -> 100%). If error rates breach 0.5% during canary shift, ALB automatically rolls back to Blue. Under primary database degradation, multi-AZ RDS initiates automatic failover to the synchronous standby in <30s. External IdP outages trigger stateless fallback using cached public keys to verify signed JWT claims locally.
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2. SERVICE LEVEL OBJECTIVES (SLOS) & ALERTING THRESHOLDS

Service Level Indicators (SLIs) are measured continuously at the ALB and application runtime layers. SLOs are evaluated over a rolling 30-day window.

Indicator (SLI)	Measurement Specification	SLO Target	30-Day Error Budget	Impact Assessment
Availability (Successful Requests)	Ratio of non-5xx responses to total valid HTTP requests at ALB: $(Total - 5XX) / Total * 100$	99.90%	0.10% total downtime allowance (~43.2 min/month)	User facing service degradation or hard outages.
Read Latency (Fast Reads)	Percentage of successful HTTP GET requests completed in < 100ms measured at Tier 1 ALB.	95.00%	5.0% of requests exceeding 100ms threshold.	UI responsiveness drop; potential client timeouts.
Write Latency (Mutations)	Percentage of successful HTTP POST/PUT/DELETE requests completed in < 300ms.	99.00%	1.0% of mutations exceeding 300ms threshold.	Workflow delays and API integration backpressure.
Database Sync Lag	Replication delay in bytes/milliseconds between Primary RDS and Multi-AZ Read Replica.	< 1000ms for 99.9% time	0.1% duration in degraded or lagging replication mode.	Stale data reads on failover or read-heavy paths.

3. MULTI-WINDOW MULTI-BURN RATE PAGING FRAMEWORK

To avoid alert fatigue and catch severe incidents rapidly, alerting is governed by Google SRE burn-rate thresholds based on the 30-day 99.9% availability error budget (Total Budget = 0.10% = 100% budget consumption at budget exhaust rate).

Severity Tier	Burn Rate Target	Short Window (Check)	Long Window (Reset)	Paging Protocol & Action
P1 - CRITICAL	14.4x Speed (2% budget consumed in 1 hour)	5 minutes (Error rate > 1.44%)	1 hour (Error rate > 1.44%)	Immediate PagerDuty Page to Primary On-Call. Auto-opens incident bridge.
P2 - HIGH	6.0x Speed (5% budget consumed in 6 hours)	30 minutes (Error rate > 0.60%)	6 hours (Error rate > 0.60%)	PagerDuty Page to On-Call engineer. Ticket opened in SRE queue.
P3 - WARNING	1.0x Speed (10% budget consumed in 3 days)	2 hours (Error rate > 0.10%)	3 days (Error rate > 0.10%)	Slack Notification (#ops-alerts). Ticket assigned for business-day resolution.

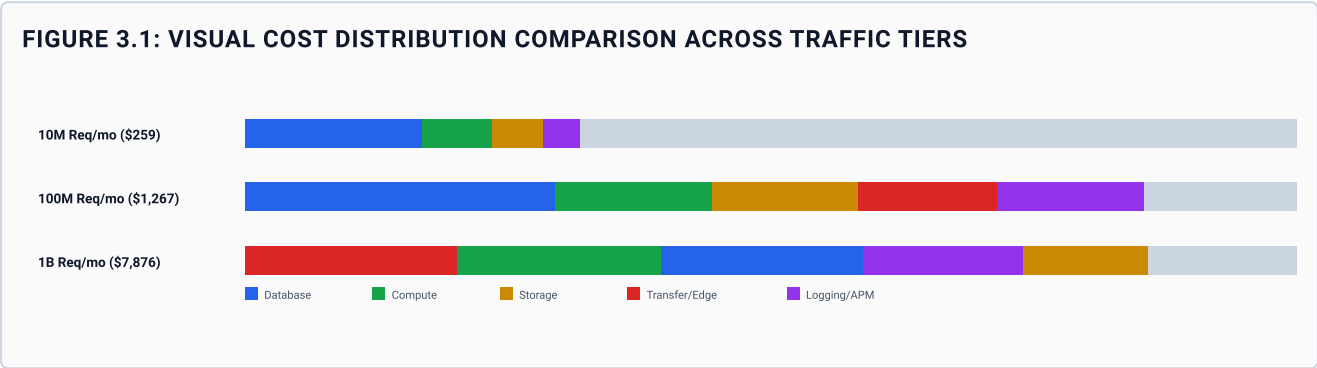
4. REAL-TIME SLO & ERROR BUDGET DASHBOARD MODEL

Availability SLO 99.94% TARGET: 99.90% (ROLLING 30D) Status: HEALTHY (+0.04% margin)	Remaining Error Budget 58.3% 25.2 MIN REMAINING OF 43.2 MIN Burn Rate: 0.8x (Nominal)	P95 Latency (Read) 42 ms TARGET: < 100 MS (95%) Headroom: 58 ms
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5. MONTHLY COST ESTIMATES ACROSS TRAFFIC TIERS

Financial projections modeled on AWS US-East-1 pricing standards across three load profiles. All tiers assume a 1:10 read-to-write ratio, 2KB payload size, and 30-day retention for operational logs.

Category / Resource	Low Scale (10M req/mo ~4 rps)	Medium Scale (100M req/mo ~40 rps)	High Scale (1B req/mo ~400 rps)
Compute Tier (ECS Fargate Tasks)	\$48.00 (2 Tasks: 0.5 vCPU, 1GB RAM)	\$192.00 (4 Tasks: 1 vCPU, 2GB RAM)	\$1,536.00 (16 Tasks: 2 vCPU, 4GB RAM)
Database Tier (RDS PostgreSQL Multi-AZ)	\$120.00 (db.t4g.medium Multi-AZ)	\$380.00 (db.r6g.xlarge Multi-AZ)	\$1,520.00 (db.r6g.4xlarge Multi-AZ)
Storage Tier (gp3 SSD + Redis Cache)	\$35.00 (50GB SSD + ElastiCache t4g.micro)	\$180.00 (300GB SSD + ElastiCache r6g.large)	\$950.00 (2TB SSD + ElastiCache cluster 3-node)
Data Transfer & Edge (CloudFront + Egress)	\$20.00 (20GB Out, CloudFront Free Tier)	\$170.00 (200GB Egress + WAF Rules)	\$1,600.00 (2TB Egress + Advanced WAF Shield)
Logging & Observability (CloudWatch + Datadog)	\$25.00 (10GB logs ingested)	\$180.00 (100GB logs ingested + metrics)	\$1,200.00 (1TB logs + APM Tracing)
Backup & Disaster Recovery (AWS Backup Snapshots + S3)	\$12.00 (Daily snapshots, 100GB total)	\$65.00 (PITR 7 days + cross-region snap)	\$350.00 (PITR 35 days + cross-region storage)
Support Plan Tier (AWS Enterprise/Business)	\$29.00 (Developer Plan fixed)	\$100.00 (Business Support - 10% min)	\$720.00 (Business Support scaled 7%)
TOTAL ESTIMATED MONTHLY COST	\$259.00 / mo	\$1,267.00 / mo	\$7,876.00 / mo



6. ARCHITECTURE DECISION RECORDS (ADRS)

ADR-001: Geographic Regional Deployment Strategy

APPROVED

Context: Service requires high availability with protection against data center level failures while balancing data consistency requirements.

Decision: Deploy in a Primary Multi-AZ Active-Passive setup in AWS us-east-1 (3 Availability Zones) with cross-region asynchronous read-replica replication to us-west-2 for disaster recovery.

Consequences: Ensures zero single-point-of-failure within primary region. Regional failover to us-west-2 requires manual DNS cutover via Route53 arc routing control.

ADR-002: Disaster Recovery Objectives (RTO & RPO)

APPROVED

Context: Business operations require strict boundaries on acceptable data loss and downtime during catastrophic outages.

Decision: Establish continuous Point-In-Time-Recovery (PITR) with automated WAL archiving to S3.

- **Recovery Time Objective (RTO):** < 30 minutes for cross-region failover; < 60 seconds for intra-region Multi-AZ database failover.

- **Recovery Point Objective (RPO):** < 5 seconds (Max asynchronous cross-region WAL lag).

Consequences: Requires continuous automated replication monitoring and quarterly regional disaster recovery failover dry-runs.

ADR-003: Secrets Management & Infrastructure Isolation

APPROVED

Context: Hardcoded credentials or unrotated secrets pose severe security risks to application data.

Decision: Store all database credentials, API keys, and certificates in AWS Secrets Manager with KMS Customer Managed Keys (CMK).

- Application containers fetch secrets dynamically at startup into ephemeral memory via IAM task roles.
- Enable mandatory 30-day automated rotation on database master credentials using AWS Lambda rotators.

Consequences: Zero plain-text credentials in Git repos or environment configurations. Application code must gracefully reconnect upon credential rotation events.

ADR-004: Least-Privilege IAM Access Control Paradigm

APPROVED

Context: Over-privileged identity roles increase the blast radius of container compromises or developer credential leaks.

Decision: Implement strict AWS IAM Boundary Policies:

- **Runtime ECS Task Role:** Granted execution rights *only* to specific KMS Key IDs, dedicated Secrets Manager ARNs, and target PostgreSQL port (5432) Security Group boundaries. No S3/Console access.
- **Human Operators:** No persistent SSH or direct DB passwords. Access enforced via AWS Systems Manager (SSM) Session Manager with IAM condition keys, fully audited in CloudTrail.

Consequences: High security baseline; administrative actions require temporary elevation via IAM Identity Center (SSO) with MFA.